Environmental Science & Technology

pubs.acs.org/est

Article

Rapid Redox Cycling of Fe(II)/Fe(III) in Microdroplets during Iron–Citric Acid Photochemistry

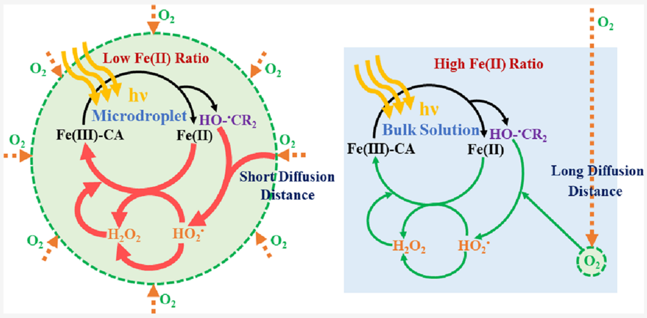
Jinzhao Wang, Di Huang, Fengxia Chen, Jianhua Chen, Hongyu Jiang, Yifan Zhu, Chuncheng Chen,* and Jincai Zhao

Cite This: Environ. Sci. Technol. 2023, 57, 4434–4442

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Fe(III) and carboxylic acids are common compositions in atmospheric microdroplet systems like clouds, fogs, and aerosols. Although photochemical processes of Fe(III)–carboxylate complexes have been extensively studied in bulk aqueous solution, relevant information on the dynamic microdroplet system, which may be largely different from the bulk phase, is rare. With the help of the custom-made ultrasonic-based dynamic microdroplet photochemical system, this study examines the photochemical process of Fe(III)–citric acid complexes in microdroplets for the first time. We find that when the degradation extent of citric acid is similar between the microdroplet system and the bulk solution, the significantly lower Fe(II) ratio is present in microdroplet samples due to the rapider reoxidation of photogenerated Fe(II). However, by replacing citric acid with benzoic acid, no much difference in the Fe(II) ratio between microdroplets and bulk solution is observed, which indicates distinct reoxidation pathways of Fe(II). Moreover, the presence of $^{\bullet}$OH scavenger, namely, methanol, greatly accelerates the reoxidation of photogenerated Fe(II) in both citric acid and benzoic acid situations. Further experiments reveal that the high availability of O_2 and the citric acid- or methanol-derived carbon-centered radicals are responsible for the rapider reoxidation of Fe(II) in iron–citric acid microdroplets by prolonging the length of $HO_2^{\bullet}$- and H_2O_2-involved radical reaction chains. The results in this study may provide a new understanding about iron–citric acid photochemistry in atmospheric liquid particles, which can further influence the photoactivity of particles and the formation of secondary organic aerosols.



KEYWORDS: *dynamic microdroplets, radical chain reactions, atmospheric photochemistry, reoxidation of Fe(II), O_2 availability*

■ INTRODUCTION

Iron is ubiquitous and abundant in natural waters and atmospheric waters, with various dissolved concentrations from several nM to tens of μM.[1−3] The bioavailability, the environmental effects, and the fate of iron species depend on their solubility and oxidation state, which would be strongly influenced by their photochemistry and complexation.[4,5] Dissolved iron exists as two oxidation states, namely, Fe(II) and Fe(III). Photochemical cycling of iron in the daytime may alter the ratio of Fe(II)/Fe(III), resulting in the higher concentration of Fe(II).[6,7] On the other hand, due to high photochemical activity and quantum yields of some Fe(III)–carboxylate complexes, the coordination of iron with low-molecular-weight carboxylic acids such as oxalic acid and citric acid (CA), which are also abundant and commonly coexistent with dissolved iron ions, can dramatically impact the photochemical redox cycling between Fe(III) and Fe(II) in the bulk aqueous phase.[8−11]

Besides existing in bulk natural waters, iron and low-molecular-weight carboxylic acids are also detected widely in atmospheric liquid particles such as clouds, fogs, and aerosols.[12,13] These particles usually have diameters in the range of 0.1 to 100 μm with a much larger specific surface area than the bulk ones.[14] Reaction behaviors in microdroplets have been reported to be much different from the bulk aqueous phase. For example, microdroplets can accelerate chemical reactions such as the Fenton reaction and oxidation of glycolaldehyde (up to three orders of magnitude).[15,16] Some reactions that cannot occur in the bulk phase become able to spontaneously happen in microdroplets.[17−21] However, current studies about photochemistry of iron as well as its complexes with carboxylic acids still focus mainly on the bulk aqueous phase.[22,23] The corresponding processes in the atmosphere-relevant droplet phase are much less understood. Recent studies on single droplets by Raman spectroscopy

Received: October 25, 2022
Revised: February 16, 2023
Accepted: February 27, 2023
Published: March 8, 2023

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

showed that the photolysis reaction of the Fe(III)−oxalate complex in a droplet was much faster than in bulk solution.[24] The degradation of organic matters such as methyl orange in the presence of the Fe(III)−oxalate complex was markedly enhanced in microdroplets, and the enhancement became more significant with the decrease in the size of the droplets from 2000 to 100 μm.[25] However, in these single-droplet studies, the droplet is deposited on the hydrophobic substrates, with a large diameter of hundreds of micrometers, and a much higher concentration of reactants than the ambient concentration has to be used due to the low detection limitation of Raman spectroscopy.[24] Relevant understanding of photochemistry in dynamic microdroplets, which are more like real clouds and fogs, is still lacking.

CA is a well-known important composition both in natural waters and atmospheric organic aerosol particles,[1,26] usually recognized as a simple proxy for humic acid. CA contains an α-hydroxyl and three carboxyl groups in favor of forming various complexations with Fe(III) in aqueous solution over a wide range of pH values from acid to neutral, avoiding precipitation.[10,27] Because of the strong complexation of CA with Fe(III), the photoinduced ligand-to-metal charge transfer (LMCT) process occurs efficiently between Fe(III) and CA, leading to the formation of Fe(II) and primary organic radical ($CA^{\bullet}$). Subsequently, $CA^{\bullet}$ undergoes fast decarboxylation to produce carbon-centered radical (CCR), namely, $HO-{}^{\bullet}CR_2$ ($R = -CH_2COOH$). The formed CCR could further react with O_2, producing a series of reactive oxygen species (ROS), which may significantly influence the valence state of iron. For example, in the viscous aerosol particles containing Fe(III) and citrate, due to the formation of the anoxic environment, photogenerated Fe(II) was accumulated in the interior of particles.[12] Photochemistry of Fe(III)−CA complexes in bulk solution is considered to play an active role in the degradation of a variety of organic and inorganic substances and has been widely investigated.[10,28−35] In these systems, ROS and high concentrations of Fe(II) were usually observed. However, in microdroplet systems, photochemistry of Fe(III)−CA complexes, such as the role of CCR and change in the oxidation state of iron, is not clearly defined.

In this study, to simulate the dynamic characteristics and low reactant concentrations of microdroplets in atmospheric liquid particles, we designed an ultrasonic-based dynamic microdroplet photochemical system, and the photochemical reactions of Fe(III)−CA complexes were examined by this microdroplet system, in comparison with those in the bulk phase. It is found that, in the Fe(III)−CA system, the Fe(II) ratio ($R_{Fe(II)}$) during the photochemical reaction in microdroplets is significantly lower than that in the bulk solution, at the similar consumption of CA regardless of the experimental conditions. Further photochemistry investigations on the Fe(III)−benzoic acid system and methanol (MeOH) presence system help us to shed light on the reason of lower $R_{Fe(II)}$ in Fe(III)−CA microdroplet samples. In addition, the difference on $R_{Fe(II)}$ in the Fe(III)−CA system between microdroplets and bulk solution is found to be largely relevant to the distinct availability of O_2 and the corresponding length of the radical reaction chain. These results contribute to a better understanding on the real photochemical process for the iron−CA complex in the natural microdroplet systems like clouds and fogs.

■ EXPERIMENTAL SECTION

Chemicals. Iron(III) perchlorate hydrate ($Fe(ClO_4)_3 \cdot xH_2O$, reagent grade, Aladdin, as the ferric source), iron perchlorate hydrate ($Fe(ClO_4)_2 \cdot xH_2O$, reagent grade, Aladdin), perchloric acid ($HClO_4$, 72%, Aladdin), citric acid anhydrous (CA, >99.5%, Alfa Aesar), benzoic acid (BA, >99.5%, Alfa Aesar), acetic acid ammonium salt (CH_3COONH_4, >98% for HPLC, Acros), sodium dihydrogen phosphate dihydrate ($NaH_2PO_4 \cdot 2H_2O$, AR, SINOPHARM), phosphoric acid (H_3PO_4, AR, SINOPHARM), hydroxylamine hydrochloride ($NH_2OH \cdot HCl$, 98%, Innochem), 1,10-phenanthroline (97%, Innochem), hydrochloric acid (AR, SINOPHARM), ammonium fluoride (NH_4F, >98%, Alfa Aesar), sodium acetate anhydrous (>99%, Innochem), acetic acid (AR, Concord Technology), ammonium iron(II) sulfate hexahydrate ($(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$, >99.997%, Sigma-Aldrich), sulfuric acid (H_2SO_4, MOS, >96%, Beijing Institute of Chemical Reagents), catalase ($>2 \times 10^5$ unit/g, Innochem), and methanol (MeOH, >99%, Beijing Chemical Works) were used as received without further purification. N_2 (>99.999%, v/v, Beijing Nanfei Industry) and O_2 (>99.999%, v/v, Beijing Nanfei Industry) were mixed by different ratios and kept at a constant flow rate of 100 standard cubic centimeters per minute (sccm). All solutions were prepared using 18 MΩ·cm resistivity ultrapure water coming from the Milli-Q ultrapure water system.

Photochemical Devices. A custom-made reactor was used to satisfy the need of investigation on dynamic microdroplet photochemistry. As depicted in Figure S1, this reactor mainly consists of a commercial ultrasonic sprayer and an about 157 mL double-walled quartz photoreactor (50 mm i.d. and 8 cm high). During photoreactions, continuously updated dynamic microdroplets were produced from the ultrasonic sprayer. Unless otherwise specified, the pH value of the newly prepared initial solutions used for microdroplet generation was adjusted to pH = 2.8 ± 0.2, which is in the range of the reported pH values of the real cloud water.[36] Measurements of the droplet-size distribution were conducted in Beijing QingXi Technology Research Institute. The diameter of these liquid particles is mainly in the range of 1 to 30 μm and centers around 9 μm, independent of the composition of solution (Figure S2). Therefore, microdroplets produced by this device give a well proxy to the real clouds and fogs, whose size is in the nanometer to micrometer range. Then, bunches of aforementioned microdroplets like clouds and fogs, through a round hole (10 mm i.d.) in the middle of the droplet storage trough (made of polytetrafluoroethylene), swarmed into the photoreactor and simultaneously suffered the light irradiation. The atmosphere in the photoreactor was controlled by introducing different gases, and the inner temperature of the photoreactor was kept at 25 °C by circulating water cooling device. After the reaction, microdroplets were condensed into the hydrophobic storage trough and collected for analysis after condensed enough amount. As shown in Figure S3a, microdroplet samples collected after different irradiation times (from 10 to 30 min) nearly have the similar $R_{Fe(II)}$ and the degradation degree of CA. Therefore, the change of irradiation time only changes the collected volume of samples but has no significant effect on the extent of the photochemical reaction in microdroplet samples. The typical irradiation time used was 15 min since sufficient samples can be obtained at that time. It is noteworthy that in the

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

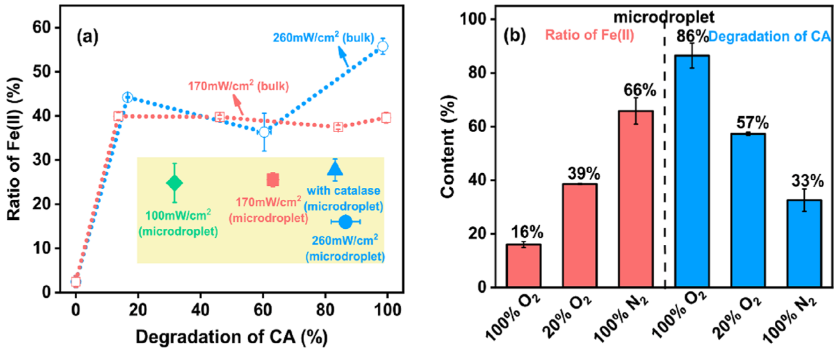


Figure 1. (a) Comparison between the Fe(III)−CA microdroplet system (solid points) and the Fe(III)−CA bulk phase (hollow points) at different light intensities under O_2 (100%) atmosphere. (b) Effect of the O_2 partial pressure on the Fe(II) ratio and CA degradation in the Fe(III)−CA microdroplet system at I = 260 mW/cm^2. The total flow rate of gas was kept at 100 sccm, and the total pressure in each reactor was 1 atm. The error bars represent one standard deviation from at least double experiments. $[\text{Fe(III)}]_0$ = 80 μM, $[\text{CA}]_0$ = 500 μM, pH = 2.8 ± 0.2.

microdroplet experiments of different aeration conditions, we just adjust the gas-phase composition but keep the atomization process unchanged. To ensure the alternation of compositions in the microdroplet system primarily due to the photochemical process, the blank reaction without irradiation was conducted (see Figure S3b). Compared with the solution before spray, the change of $R_{\text{Fe(II)}}$ and CA concentration in microdroplet samples are all insignificant, which indicates that the effects of droplet evaporation and condensation in the dark are negligible. Hence, the ultrasonic-based dynamic microdroplet photochemical system used here is appropriate to study microdroplet photochemical reactions. To examine the photochemical reactions in bulk solution, 50 mL of solution was introduced into a 60 mL double-layer cylindrical quartz reactor (12 mm i.d.), keeping continuously stirred at 25 °C and gas bubbling into solutions throughout irradiation. The samples (2 mL) were taken out at given time intervals to quantify both the concentration of ferrous ion and residual carboxylic acid. All the reactors above were washed three times by absolute ethanol and deionized water respectively before utilization.

A xenon lamp (CEL-HXF 300, Beijing Ceaulight, China) equipped with a UV reflector film was used for both the microdroplet and bulk reactions. The optical power density at the center of the reactors used for microdroplet and bulk systems was measured by an optical power meter (Beijing Ceaulight, China). Figure S4 shows the emission spectrum of the light source and the UV−vis absorption spectrum of three iron-containing solutions. The emission spectrum of the light source spreads over the wavelength range of 350−500 nm, which is able to induce the LMCT process in Fe(III)−CA and Fe(III)−BA systems. In addition, the radiation in this wavelength range is also abundant in the lower troposphere.

Analytical Methods. Concentrations of CA and BA and their degradation products in the liquid phase were determined by a high-performance liquid chromatography (HPLC) system equipped with a UV detector (1200 Series, Agilent, USA). A Dikma Spursil C18 reversed-phase column (250 × 4.6 mm, 5 μm) was used for separation. The detailed HPLC methods and the calculation of the degradation ratio of two carboxylic acids are shown in Text S2. The volatile products (oxygenated VOCs) during the photoreaction in microdroplets were detected by PTR-ToF-MS 1000 (proton-transfer reaction time-of-flight mass spectrometer, Ionicon, Austria). A colorimetric method to measure the concentrations of total iron and ferrous ion was previously described by Wang et al.[37] 1,10-Phenanthroline was used as the ligand to coordinate with Fe(II), forming a specific colored complex. After 15 min of standing in the dark, the absorbance at 510 nm of the chromogenic agent and sample mixture was recorded by a UV−vis spectrophotometer (U-3900, Hitachi, Japan) (details of this analysis process are shown in Text S3). The calibration curve of Fe(II) concentration is shown in Figure S5a, and the linear range of this method meets the experimental needs. The presence of CA or BA has little effect on the determination of Fe concentration as well as on the calibration curve of Fe(II) (Figure S5b).

RESULTS AND DISCUSSION

Comparison between Photochemical Reactions of the Fe(III)−CA Complex in Bulk Solution and in the Microdroplet System. First, the photochemical reaction of Fe(III)−CA complexes was performed in bulk solution. As shown in Figure S6, accompanying with the degradation of CA, Fe(II) is formed rapidly upon the irradiation and reaches a relative steady state within 30 s. In the photochemical experiments of microdroplets, as mentioned before, microdroplets are continuously produced from the ultrasonic sprayer and deposited onto the bottom of the reactor after undergoing irradiation. The final samples are collected from the bottom after 15 min of irradiation. In this way, it is difficult to obtain the kinetic factors for the photoreaction of a single microdroplet as in bulk solution. Moreover, the completely different optical properties of the microdroplet and bulk phase lead to the light absorption between them not comparable. Therefore, the similar degradation ratio of CA is selected as a reference to compare the $R_{\text{Fe(II)}}$ in total iron in the microdroplet and bulk solution, which can reasonably reflect the relative reoxidation rate of photogenerated Fe(II) between these two systems. Figure 1a shows the $R_{\text{Fe(II)}}$ and the degradation of CA during the photochemistry of Fe(III)−CA complexes in the microdroplet system and in bulk solution at different light intensities. In the microdroplet system, the degradation of CA increases from 32% at I = 100 mW/cm^2 to 86% at I = 260 mW/cm^2, indicating that the light intensity markedly influences the photochemical reaction in the microdroplet system. The $R_{\text{Fe(II)}}$ is 26% at I = 170 mW/cm^2 and becomes 16% at I = 260 mW/cm^2. For the bulk solution, the $R_{\text{Fe(II)}}$ remains larger than 40% as long as CA is not depleted. The most notable influence of light intensity during bulk phase photochemistry is on the $R_{\text{Fe(II)}}$ just before the depletion of CA. At this point, the $R_{\text{Fe(II)}}$

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

is 56% at I = 260 mW/cm^2, while it becomes 40% at I = 170 mW/cm^2. To further test the difference of $R_{Fe(II)}$ between microdroplet and bulk systems, we conducted the photochemical experiments of the Fe(III)−CA system at a higher pH. As shown in Figure S7, although the slower degradation rate of CA and the lower $R_{Fe(II)}$ are observed at pH = 6.2 in bulk solution, the microdroplet sample also has significantly lower $R_{Fe(II)}$ than the bulk phase, in line with the case of low pH. In addition, different effects of the initial concentration ratio of Fe(III) to CA on CA degradation and iron cycling are also observed between the microdroplet and bulk systems. As shown in Figure S8, in bulk solution, it is found that, at the earlier stage of irradiation (before 150 s), the degradation of CA (ΔC_{CA} ∼220 μM) exhibits little difference between the systems of Fe(III):CA = 80/500 and 80/320. By contrast, in microdroplets, the degradation of CA in the Fe(III):CA = 80/500 system (ΔC_{CA} = 315 μM) is significantly rapider than in the case of 80/320 (ΔC_{CA} = 236 μM). Moreover, at both Fe(III):CA, $R_{Fe(II)}$ is much lower than that in the bulk solution at the similar degradation degree of CA (Figure S8c). Except for changing the concentration of CA, a lower initial concentration ratio of Fe(III) to CA was obtained by decreasing the concentration of iron, and the data points for $R_{Fe(II)}$ from the microdroplet system are also located far below the curve of the bulk solution (Figure S9). It is evident that, regardless of light intensities, pH, and concentration ratio of reactants, the $R_{Fe(II)}$ obtained in the microdroplet system is much lower than that in the bulk system at the similar degradation degree of CA.

We further investigated the influence of O_2 partial pressure (P_{O_2}) on the $R_{Fe(II)}$ and the degradation of CA in the photochemistry of the Fe(III)−CA microdroplet system. As shown in Figure 1b, with the decrease in P_{O_2}, the $R_{Fe(II)}$ is remarkably enhanced, while the degradation of CA is largely suppressed, which indicates that O_2 plays an important role in the photochemistry of Fe(III)−CA in the microdroplet systems. The rather high $R_{Fe(II)}$ (up to 66%) obtained from the pure N_2 atmosphere demonstrates that Fe(II) can be generated efficiently in the microdroplet system, and thus the lower $R_{Fe(II)}$ for the microdroplet system in the presence of O_2 (Figure 1a) should stem from the O_2-involved reoxidation process of Fe(II) rather than from its slow photoinduced formation.

Comparison between Photochemical Reactions of Fe(III)−CA and Fe(III)−BA Systems. BA is a small aromatic acid from biomass burning and is ample in cloud water and rainwater of polluted regions.[38,39] Different from CA, BA has weak interaction with Fe(III), as shown by unchanged UV−vis absorption spectra of Fe(III)−BA solution relative to Fe(III) alone (Figure S4). An examination on the $R_{Fe(II)}$ during Fe(III)−BA photochemistry in the bulk system shows that, with the degradation of BA, the $R_{Fe(II)}$ gradually increases, and it can accumulate up to 89% after 80% of BA is degraded (Figure 2). At a lower light intensity, although the slower BA degradation rate and lower $R_{Fe(II)}$ are observed in the bulk solution (Figure S10a), the similar $R_{Fe(II)}$ is obtained at the similar degradation degree of BA under both light intensities. Moreover, the difference in the $R_{Fe(II)}$ at the similar degradation extent of BA is not significant between the bulk solution and the microdroplet system regardless of light intensity (Figure S10b), which is largely distinct from the Fe(III)−CA situation (Figure 1a). In the microdroplet system,

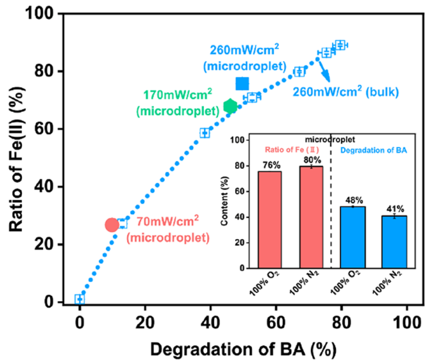


Figure 2. Comparison between the Fe(III)−BA microdroplet system (solid points) and the Fe(III)−BA bulk phase (hollow points) at different light intensities under O_2 (100%) atmosphere. Inset: Effect of the O_2 partial pressure on the Fe(II) ratio and BA degradation in the Fe(III)−BA microdroplet system. The total flow rate of gas was kept at 100 sccm, and the total pressure in each reactor was 1 atm. The error bars represent one standard deviation from at least double experiments. $[Fe(III)]_0$ = 80 μM, $[BA]_0$ = 80 μM, pH = 2.8 ± 0.2.

both the $R_{Fe(II)}$ and the degradation of BA are markedly enhanced with the increase of light intensity. When the light intensity increases from 70 to 260 mW/cm^2, the $R_{Fe(II)}$ increases from 27 to about 76% (solid points in Figure 2). These observations are in sharp contrast with the Fe(III)−CA case, where the $R_{Fe(II)}$ remains at a quite low level and does not alter a lot with light intensity. Moreover, in the absence of O_2, both the $R_{Fe(II)}$ and the degradation of BA are not changed much, relative to that at 100% O_2 partial pressure (Figure 2 inset), which is also different from the large O_2 dependence of $R_{Fe(II)}$ in the Fe(III)−CA system (Figure 1b).

Based on the species distribution simulation by Visual MINTEQ, the dominant Fe(III) complex is $FeOH^{2+}$ (76%), and the complexation of Fe(III) with BA is very weak under our experimental conditions. Accordingly, the photochemical reaction in the Fe(III)−BA system is initiated by the photolysis of the $FeOH^{2+}$ complex, producing $^{\bullet}OH$ radicals and Fe(II) (eq 1). The degradation of BA stems from the attack of $^{\bullet}OH$ radicals and the formation of $^{\bullet}OH$ adduct (BA−$^{\bullet}$OH).[40,41] The further reaction of BA−$^{\bullet}$OH with O_2 will lead to the formation of bicyclic peroxyl radicals, besides $HO_2^{\bullet}$ radicals. As found in the toluene oxidation by $^{\bullet}OH$ in water, the bicyclic peroxyl pathway was dominant in the oxidation process, and the formation of $HO_2^{\bullet}$ only accounted for the minor fraction.[42] However, in the Fe(III)−CA system, different from the case in the Fe(III)−BA system, the formation of $HO_2^{\bullet}$ and O=CR_2 (R = $-CH_2COOH$) is the dominant pathway for the reaction of HO−$^{\bullet}CR_2$ with O_2. The generation of $HO_2^{\bullet}$ will promote the cycling of iron and the successive production of ROS including H_2O_2 and $^{\bullet}OH$, triggering the radical chain reactions in the participation of O_2. Hence, the role of O_2 is more obvious in the Fe(III)−CA system than the Fe(III)−BA system no matter on the $R_{Fe(II)}$ or on the degradation of carboxylic acid. A detection on the oxidation products of CA shows that acetone, 3-oxoglutaric acid, and acetoacetic acid are the main products in the gas phase and in the collected liquid samples (Figure S11). It is evident that the 3-oxoglutaric acid is formed through the decarboxylation of CA, which verifies the existence of CCR (HO−$^{\bullet}CR_2$). The acetone and acetoacetic acid should originate from the secondary decarboxylation of 3-oxoglutaric acid, which suggests the occurrence of radical chain reactions.

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

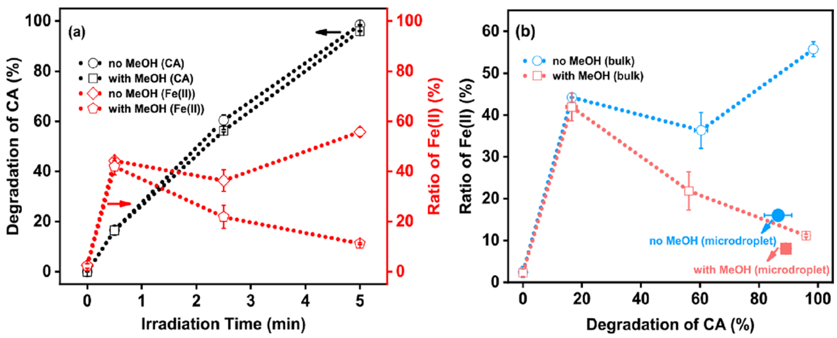


Figure 3. (a) Effect of MeOH on the CA degradation rate and Fe(II) ratio in the Fe(III)−CA bulk photochemical system. (b) Comparison between the Fe(III)−CA microdroplet system (solid points) and the Fe(III)−CA bulk phase (hollow points) with or without MeOH. $[Fe(III)]_0$ = 80 μM, $[CA]_0$ = 500 μM, $[MeOH]_0$ = 0.25 M, pH = 2.8 ± 0.2, I = 260 mW/cm^2. All photochemical experiments were conducted under O_2 (100%) atmosphere. The total flow rate of O_2 was kept at 100 sccm, and the total pressure in each reactor was 1 atm.

The necessity of the radical chain reactions to the reoxidation of photogenerated Fe(II) will be further discussed below.

$$FeOH^{2+} \xrightarrow{h\nu} Fe(II) + {}^{\bullet}OH \quad (1)$$

Oxygen-Involved Radical Chain Reactions in the Fe(III)−CA Microdroplet Photochemical System. The $^{\bullet}$OH radical has been proposed as one of the most important ROS for the degradation of various organic pollutants and the oxidation of photogenerated Fe(II) during the photochemistry of iron.[16,23,27,43] To experimentally examine the oxidation of Fe(II) by $^{\bullet}$OH (eq 2), MeOH (0.25 M), which has a weak effect on the coordination of Fe(III) but is able to react rapidly with $^{\bullet}$OH (eq 3),[6] was added into the Fe(III)−CA solution. As shown in Figure 3a, in bulk solution, the degradation rate of CA is hardly influenced by MeOH. However, with the degradation of CA, the $R_{Fe(II)}$ in the presence of MeOH becomes lower and lower and decreases to just 11% when CA is nearly depleted. By contrast, the $R_{Fe(II)}$ in solution without MeOH keeps at a relatively high level (>40%) and does not change much during overall photoreaction. In the microdroplet system (Figure 3b), the presence of MeOH also has little impact on the degradation of CA. However, a significant decrease in $R_{Fe(II)}$ (from 16 to 8%) by adding MeOH is observed. For the Fe(III)−BA bulk system (Figure S12), different from the unchanged degradation of CA in the Fe(III)−CA system (Figure 3a), the degradation of BA is completely hindered by MeOH added due to the competition of MeOH with BA for $^{\bullet}$OH. However, the reoxidation of Fe(II) is markedly enhanced relative to the case without MeOH. In the microdroplet system (Figure S13), the presence of MeOH also significantly hinders the degradation of BA, and the $R_{Fe(II)}$ is greatly lowered (from 76% without MeOH to 28% in O_2 atmosphere), indicating that the existence of MeOH can also promote the reoxidation of Fe(II) in the Fe(III)−BA microdroplet system. In MeOH presence systems, the concentration of $^{\bullet}$OH should be markedly decreased. The lower $R_{Fe(II)}$ after adding MeOH in all the above systems suggests that $^{\bullet}$OH is not the dominant active species for the reoxidation of Fe(II) in both the microdroplet and bulk systems.

Besides $^{\bullet}$OH, $HO_2^{\bullet}$ and H_2O_2 are also the possible active species for the reoxidation of Fe(II) in bulk solution.[23,44,45] To make clear the importance of $HO_2^{\bullet}$ and H_2O_2 to the reoxidation of Fe(II) in the microdroplet system, we first examined the effect of catalase, which can efficiently decompose H_2O_2 to water and oxygen and thus avoid its reaction with Fe(II) (eq 10). It is found that, in the presence of catalase, the $R_{Fe(II)}$ in the microdroplet system is obviously higher than that in the absence of catalase (blue points in Figure 1a), indicating that H_2O_2 does play an important role in the reoxidation of Fe(II). To further shed light on the role of $HO_2^{\bullet}$ on the reoxidation of Fe(II), samples collected from the photochemical reaction of microdroplets were kept in the dark to avoid the photo-formation of $HO_2^{\bullet}$ (eqs 4−7). As shown in Figure 4, when the dark reaction is carried out in the aerobic

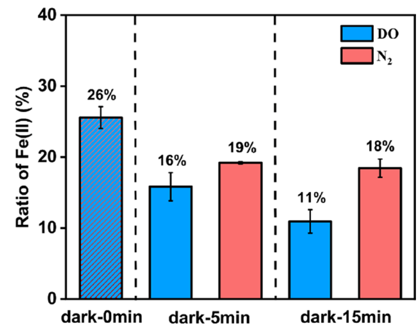


Figure 4. The dark reoxidation of photogenerated Fe(II) in microdroplet samples with DO or under the condition of N_2 bubbling. Photoreaction: $[Fe(III)]_0$ = 80 μM, $[CA]_0$ = 500 μM, pH = 2.8 ± 0.2, I = 170 mW/cm^2, 100% O_2 partial pressure. Dark reaction: with or without N_2 bubbling. The total flow rate of gas was kept at 100 sccm.

environment, the $R_{Fe(II)}$ decreases from 26 to 11% within 15 min. Since photoinduced $HO_2^{\bullet}$ has very short lifetime in water, once irradiation ceased, $HO_2^{\bullet}$ in microdroplet samples disappears immediately. Therefore, the obvious reoxidation of Fe(II) during the dark reaction is mainly initiated by long-lived oxidant H_2O_2. Compared with the aerobic environment, when N_2 is continuously bubbled to remove the dissolved oxygen (DO) from samples collected after the aerated photochemical reaction in the microdroplet system, the decrease in the $R_{Fe(II)}$ becomes less significant (from 26 to 18%). In the control experiment, for the aerated solution of Fe(II) and CA, no change in $R_{Fe(II)}$ is observed after a 30 min dark reaction in bulk solution (Figure S14), which indicates that the direct oxidation of Fe(II) by molecular oxygen (eq 13) under the present conditions is too slow to be considered. These observations suggest that DO can contribute to the H_2O_2-initiated oxidation of Fe(II) and the more possible participation of DO in Fe(II) reoxidation is through radical

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

reactions listed in eqs 11 and 12. $^{\bullet}OH$ produced by the reaction between Fe(II) and H_2O_2 can oxidize CA to organic radicals, which is subsequently able to reduce DO to $HO_2^{\bullet}$ under aerobic acidic conditions. The appearance of $HO_2^{\bullet}$ would largely promote the regeneration of H_2O_2 (eqs 8 and 9) and the reoxidation of Fe(II),[46] leading to the lower $R_{Fe(II)}$ in the presence of DO during the dark reaction (Figure 4). Hence, $HO_2^{\bullet}$ and H_2O_2 are both vital to the reoxidation of Fe(II) in the Fe(III)−CA microdroplet system.

The importance of $HO_2^{\bullet}$ to the reoxidation of Fe(II) can also be demonstrated by the effects of MeOH. In the Fe(III)−BA microdroplet system without MeOH, the P_{O_2} hardly influences the $R_{Fe(II)}$ (Figure 2 inset). By contrast, in the presence of MeOH, $R_{Fe(II)}$ is much lower under O_2 atmosphere (28%) than that in N_2 (60%; Figure S13), which exhibits a similar P_{O_2} effect to the Fe(III)−CA system (Figure 1b). In both Fe(III)−CA and Fe(III)−BA systems, the added MeOH can react with the photoinduced $^{\bullet}OH$ to form $^{\bullet}CH_2OH$ radical. Similar to the CA-derived HO−$^{\bullet}CR_2$, $^{\bullet}CH_2OH$ subsequently is able to react with O_2 to generate $HO_2^{\bullet}$, which induces the radical chain reactions and rapid oxidation of Fe(II). In the MeOH absence Fe(III)−CA bulk system, the formation of $HO_2^{\bullet}$ is possible to be gradually depressed with irradiation time, due to the rapid consumption of DO inside the bulk solution, and the untimely supplement of O_2 from the interface to the interior of the solution. To verify the presence of DO deficiency in Fe(III)−CA bulk solution during the photochemical reaction, the DO concentration measurement was conducted. As shown in Figure S15, the initial concentration of DO before irradiation is kept at a constant (31 mg/L). Upon irradiation, DO decreases continuously and drops to 22 mg/L at 8 min (CA is nearly depleted), even if pure O_2 was purged into the solution with the flow rate of 100 sccm, which confirms the long-distance diffusion limitation of O_2 in the bulk phase. However, such a deficiency of DO hardly occurs in microdroplet systems due to the short diffusion distance of O_2.[25] Since $HO_2^{\bullet}$ formed by the reaction of HO−$^{\bullet}CR_2$ and DO is essential to ensure the radical chain reactions in the Fe(III)−CA photochemical system, the deficiency of DO will seriously shorten the length of the radical reaction chain (eqs 4−10). That is the reason for the lower ratio of photogenerated Fe(II) in microdroplet systems than in bulk solution during Fe(III)−CA photochemistry.

$$Fe(II) + {}^{\bullet}OH \rightarrow Fe(III) + OH^- \quad (2)$$

$$CH_3OH + {}^{\bullet}OH \rightarrow {}^{\bullet}CH_2OH + H_2O \quad (3)$$

$$Fe(III) - CA \xrightarrow{h\nu} Fe(II) + CA^{\bullet} \quad (4)$$

$$CA^{\bullet} \rightarrow HO - {}^{\bullet}CR_2\ (R = -CH_2COOH) + CO_2 \quad (5)$$

$$HO - {}^{\bullet}CR_2 + O_2 \rightarrow O = CR_2 + HO_2^{\bullet}\ (R = -CH_2COOH) \quad (6)$$

$$O_2^{\bullet -} + H^+ \rightleftharpoons HO_2^{\bullet} \quad (7)$$

$$Fe(II) + HO_2^{\bullet} + H^+ \rightarrow Fe(III) + H_2O_2 \quad (8)$$

$$2HO_2^{\bullet} \rightarrow H_2O_2 + O_2 \quad (9)$$

$$Fe(II) + H_2O_2 \rightarrow Fe(III) + {}^{\bullet}OH + OH^- \quad (10)$$

$$^{\bullet}OH + CA \rightarrow CA^{\bullet} + H_2O \quad (11)$$

$$CA^{\bullet} + O_2 \rightarrow HO_2^{\bullet} + product \quad (12)$$

$$Fe(II) + O_2 \rightarrow Fe(III) + HO_2^{\bullet}/O_2^{\bullet -} \quad (13)$$

ATMOSPHERIC IMPLICATIONS

Microdroplet systems like clouds, fogs, and aerosols are abundant in the atmosphere and can serve as microreactors for compound conversions.[36,47,48] Under sunlit, photochemical reactions may become especially important in natural microdroplet systems. As reported, there are numerous photoactive species in atmospheric conditions.[14,49] Among them, Fe(III)−carboxylate complexes, which can experience the photoinduced LMCT process to influence the cycling of iron and the oxidation of organic matters, are at high concentrations and widely studied in the bulk phase.[22,41] In this study, we find that, comparing with the well-studied bulk phase photochemistry of Fe(III)−CA, the photochemical reaction of Fe(III)−CA complexes in microdroplets shows significant lower $R_{Fe(II)}$ at the similar degradation extent of CA. In natural waters such as rivers, lakes, and seas, the bulk reaction is dominant, but in atmospheric waters such as clouds, fogs, and aerosols, the photochemical reaction in the microdroplet or particle phase plays an important role. Our results suggest that the Fe-based photochemistry may be much different between the bulk waters and particulate waters. During the photochemical reaction of Fe(III)−CA complexes in bulk solution, the long diffusion distance from the surface to the bulk (usually several centimeters) does not favor the diffusion of O_2 into the interior of bulk solution. Accordingly, a large amount of Fe(II) can be accumulated due to the poor availability of O_2. It is also reported recently that, in the Fe(III)−CA contained aerosol particles, the viscosity is very high at low relative humidity, which seriously limits the diffusion of O_2 into the interior of particles, and also results in the accumulation of Fe(II) in the particles.[12] However, the short diffusion distance and rapid diffusion rate of O_2 in microdroplets ensure the availability of O_2 in microdroplets during Fe(III)−CA photochemistry and lead to much lower $R_{Fe(II)}$ in both O_2 partial pressures of 100 and 20% (Figure S16). Therefore, our photochemical experiments in the Fe(III)−CA microdroplet system provide a new understanding on cycling of iron when the diffusion of O_2 is no longer a restricted factor.

Previous modeling for cloud droplets suggested that Fe(II) was the dominant oxidation state of iron during the daytime due to the photoreduction of Fe(III), which was the combined results of various Fe(III)−carboxylate complexes.[50] In this study, comparing the microdroplet photochemistry occurring respectively in Fe(III)−CA and Fe(III)−BA systems, we also shed insightful light onto the specific mechanistic difference; that is, the rapider reoxidation of Fe(II) in Fe(III)−CA microdroplets is attributed to the high O_2 availability due to the CCR-involved radical reactions. These radical reactions are determined by the identity of CCR. Accordingly, different carboxylic acids may devote different radical reactions and different $R_{Fe(II)}$ in clouds. In the Fe(III)−CA microdroplet, for example, the $R_{Fe(II)}$ is always lower than 40% under aerobic conditions, but in the case of BA, the $R_{Fe(II)}$ is able to reach a quite high level of 76%. Ignoring the lower $R_{Fe(II)}$ in the Fe(III)−CA microdroplet system may cause the overestimation of Fe(II) concentration during the photochemistry

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442

of atmospheric liquid particles. In addition, due to the stronger coordination capability and photoactivity of Fe(III) than Fe(II), rapid reoxidation of photogenerated Fe(II) during Fe(III)−CA photochemistry in microdroplets is able to maintain the photochemical cycling of iron continuously and promote the production of ROS, which could provide strong oxidation capability to the tropospheric aqueous phase and may further cause the formation and loss of secondary organic aerosols like previously reported.[12,13,16,51] We also realize that the determination of photochemical kinetics in the microdroplet reaction is very useful and important. However, it is challenging for the used ultrasonic-based dynamic microdroplet photochemical reactor to access kinetic information due to the difficulty in controlling and obtaining the real photoreaction time in each microdroplet. The length-tunable microdroplet photoreactor, combining with mass spectrum detection,[52,53] may make the kinetic information detectable.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.2c07897.

> Additional experimental details, picture of the experimental device, analytical methods (HPLC, UV−vis, dissolved oxygen detection, PTR-ToF-MS, and pH), characterization of the microdroplet size, and the influences of additional experimental conditions on the photochemical reactions (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Chuncheng Chen − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China;* orcid.org/0000-0003-4034-8063; Email: ccchen@iccas.ac.cn

Authors

Jinzhao Wang − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Di Huang − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Fengxia Chen − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Jianhua Chen − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Hongyu Jiang − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Yifan Zhu − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China*

Jincai Zhao − *Beijing National Laboratory for Molecular Sciences, Key Laboratory of Photochemistry, CAS Research/Education Center for Excellence in Molecular Sciences, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, P. R. China; University of Chinese Academy of Sciences, Beijing 100049, P. R. China;* orcid.org/0000-0003-1449-4235

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.est.2c07897

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Strategic Priority Research Program of Chinese Academy of Sciences (grant no. XDB36000000), the National Natural Science Foundation of China (nos. 22136005, 22188102, 22206188, and 21827809), the "National Key R&D Program of China" (nos. 2018YFA0209302 and 2020YFA0710303), and the China Postdoctoral Science Foundation (no. 2021M703270).

■ REFERENCES

(1) Deguillaume, L.; Leriche, M.; Desboeufs, K.; Mailhot, G.; George, C.; Chaumerliac, N. Transition Metals in Atmospheric Liquid Phases: Sources, Reactivity, and Sensitive Parameters. *Chem. Rev.* **2005**, *105*, 3388−3431.

(2) Jacob, D. J. Heterogeneous chemistry and tropospheric ozone. *Atmos. Environ.* **2000**, *34*, 2131−2159.

(3) Paris, R.; Desboeufs, K. V. Effect of atmospheric organic complexation on iron-bearing dust solubility. *Atmos. Chem. Phys.* **2013**, *13*, 4895−4905.

(4) Kim, K.; Choi, W.; Hoffmann, M. R.; Yoon, H.-I.; Park, B.-K. Photoreductive Dissolution of Iron Oxides Trapped in Ice and Its Environmental Implications. *Environ. Sci. Technol.* **2010**, *44*, 4142−4148.

(5) Wei, J.; Yu, H.; Wang, Y.; Verma, V. Complexation of Iron and Copper in Ambient Particulate Matter and Its Effect on the Oxidative Potential Measured in a Surrogate Lung Fluid. *Environ. Sci. Technol.* **2019**, *53*, 1661−1671.

(6) Willey, D.; Whitehead, R. F.; Kieber, R. J.; Hardison, D. R. Oxidation of Fe(II) in Rainwater. *Environ. Sci. Technol.* **2005**, *39*, 2579−2585.

(7) Kieber, R. J.; Skrabal, S. A.; Smith, B. J.; Willey, J. D. Organic Complexation of Fe(II) and Its Impact on the Redox Cycling of Iron in Rain. *Environ. Sci. Technol.* **2005**, *39*, 1576−1583.

(8) Weller, C.; Horn, S.; Herrmann, H. Effects of Fe(III)-concentration, speciation, excitation-wavelength and light intensity on the quantum yield of iron(III)-oxalato complex photolysis. *J. Photochem. Photobiol., A* **2013**, *255*, 41−49.

(9) Rodríguez, E. M.; Núñez, B.; Fernández, G.; Beltrán, F. J. Effects of some carboxylic acids on the Fe(III)/UVA photocatalytic oxidation of muconic acid in water. *Appl. Catal., B* **2009**, *89*, 214−222.

(10) Feng, X.; Chen, Y.; Fang, Y.; Wang, X.; Wang, Z.; Tao, T.; Zuo, Y. Photodegradation of parabens by Fe(III)-citrate complexes at circumneutral pH: Matrix effect and reaction mechanism. *Sci. Total Environ.* **2014**, *472*, 130−136.

(11) Wu, B.; Zhang, G.; Zhang, L.; Song, X.; Zhang, S.; Zhu, B.-Z. Key factors in the ligand effects on the photo redox cycling of aqueous iron species. *Geochim. Cosmochim. Acta* **2020**, *281*, 1−11.

(12) Alpert, P. A.; Dou, J.; Corral Arroyo, P.; Schneider, F.; Xto, J.; Luo, B.; Peter, T.; Huthwelker, T.; Borca, C. N.; Henzler, K. D.; Schaefer, T.; Herrmann, H.; Raabe, J.; Watts, B.; Krieger, U. K.; Ammann, M. Photolytic radical persistence due to anoxia in viscous aerosol particles. *Nat. Commun.* **2021**, *12*, 1769.

(13) Dou, J.; Alpert, P. A.; Corral Arroyo, P.; Luo, B.; Schneider, F.; Xto, J.; Huthwelker, T.; Borca, C. N.; Henzler, K. D.; Raabe, J.; Watts, B.; Herrmann, H.; Peter, T.; Ammann, M.; Krieger, U. K. Photochemical degradation of iron(III) citrate/citric acid aerosol quantified with the combination of three complementary experimental techniques and a kinetic process model. *Atmos. Chem. Phys.* **2021**, *21*, 315−338.

(14) Herrmann, H.; Schaefer, T.; Tilgner, A.; Styler, S. A.; Weller, C.; Teich, M.; Otto, T. Tropospheric Aqueous-Phase Chemistry: Kinetics, Mechanisms, and Its Coupling to a Changing Gas Phase. *Chem. Rev.* **2015**, *115*, 4259−4334.

(15) Enami, S.; Sakamoto, Y.; Colussi Agustín, J. Fenton chemistry at aqueous interfaces. *Proc. Natl. Acad. Sci. U. S. A.* **2014**, *111*, 623−628.

(16) Thomas, D. A.; Coggon, M. M.; Lignell, H.; Schilling, K. A.; Zhang, X.; Schwantes, R. H.; Flagan, R. C.; Seinfeld, J. H.; Beauchamp, J. L. Real-Time Studies of Iron Oxalate-Mediated Oxidation of Glycolaldehyde as a Model for Photochemical Aging of Aqueous Tropospheric Aerosols. *Environ. Sci. Technol.* **2016**, *50*, 12241−12249.

(17) Lee Jae, K.; Walker Katherine, L.; Han Hyun, S.; Kang, J.; Prinz Fritz, B.; Waymouth Robert, M.; Nam Hong, G.; Zare Richard, N. Spontaneous generation of hydrogen peroxide from aqueous microdroplets. *Proc. Natl. Acad. Sci. U. S. A.* **2019**, *116*, 19294−19298.

(18) Nie, H.; Wei, Z.; Qiu, L.; Chen, X.; Holden, D. T.; Cooks, R. G. High-yield gram-scale organic synthesis using accelerated microdroplet/thin film reactions with solvent recycling. *Chem. Sci.* **2020**, *11*, 2356−2361.

(19) Gong, C.; Li, D.; Li, X.; Zhang, D.; Xing, D.; Zhao, L.; Yuan, X.; Zhang, X. Spontaneous Reduction-Induced Degradation of Viologen Compounds in Water Microdroplets and Its Inhibition by Host−Guest Complexation. *J. Am. Chem. Soc.* **2022**, *144*, 3510−3516.

(20) Lee, J. K.; Samanta, D.; Nam, H. G.; Zare, R. N. Micrometer-Sized Water Droplets Induce Spontaneous Reduction. *J. Am. Chem. Soc.* **2019**, *141*, 10585−10589.

(21) Qiu, J.; Ishizuka, S.; Tonokura, K.; Enami, S. Interfacial vs Bulk Ozonolysis of Nerolidol. *Environ. Sci. Technol.* **2019**, *53*, 5750−5757.

(22) Clarizia, L.; Russo, D.; Di Somma, I.; Marotta, R.; Andreozzi, R. Homogeneous photo-Fenton processes at near neutral pH: A review. *Appl. Catal., B* **2017**, *209*, 358−371.

(23) Wang, Z.; Liu, J. New insight into photochemical oxidation of Fe(II): The roles of Fe(III) and reactive oxygen species. *Catal. Today* **2014**, *224*, 244−250.

(24) Wang, Y.; Brigante, M.; Mailhot, G.; Talaga, D.; Wu, Y.; Dong, W.; Sobanska, S. Toward a better understanding of ferric-oxalate complex photolysis: The role of the aqueous/air interface of droplet. *Chemosphere* **2022**, *289*, No. 133127.

(25) Li, K.; Gong, K.; Liu, J.; Ohnoutek, L.; Ao, J.; Liu, Y.; Chen, X.; Xu, G.; Ruan, X.; Cheng, H.; Han, J.; Sui, G.; Ji, M.; Valev, V. K.; Zhang, L. Significantly accelerated photochemical and photocatalytic reactions in microdroplets. *Cell Rep. Phys. Sci.* **2022**, *3*, No. 100917.

(26) Reid, J. P.; Bertram, A. K.; Topping, D. O.; Laskin, A.; Martin, S. T.; Petters, M. D.; Pope, F. D.; Rovelli, G. The viscosity of atmospherically relevant organic particles. *Nat. Commun.* **2018**, *9*, 956.

(27) Chen, Y.; Liu, Z.; Wang, Z.; Xue, M.; Zhu, X.; Tao, T. Photodegradation of propranolol by Fe(III)−citrate complexes: Kinetics, mechanism and effect of environmental media. *J. Hazard. Mater.* **2011**, *194*, 202−208.

(28) Ou, X.; Quan, X.; Chen, S.; Zhang, F.; Zhao, Y. Photocatalytic reaction by Fe(III)−citrate complex and its effect on the photodegradation of atrazine in aqueous solution. *J. Photochem. Photobiol., A* **2008**, *197*, 382−388.

(29) Abida, O.; Kolar, M.; Jirkovsky, J.; Mailhot, G. Degradation of 4-chlorophenol in aqueous solution photoinduced by Fe(III)−citrate complex. *Photochem. Photobiol. Sci.* **2012**, *11*, 794−802.

(30) Pozdnyakov, I. P.; Kolomeets, A. V.; Plyusnin, V. F.; Melnikov, A. A.; Kompanets, V. O.; Chekalin, S. V.; Tkachenko, N.; Lemmetyinen, H. Photophysics of Fe(III)−tartrate and Fe(III)−citrate complexes in aqueous solutions. *Chem. Phys. Lett.* **2012**, *530*, 45−48.

(31) Abrahamson, H. B.; Rezvani, A. B.; Brushmiller, J. G. Photochemical and spectroscopic studies of complexes, of iron(III) with citric acid and other carboxylic acids. *Inorg. Chim. Acta* **1994**, *226*, 117−127.

(32) Baba, Y.; Yatagai, T.; Harada, T.; Kawase, Y. Hydroxyl radical generation in the photo-Fenton process: Effects of carboxylic acids on iron redox cycling. *Chem. Eng. J.* **2015**, *277*, 229−241.

(33) Zhang, C.; Wang, L.; Wu, F.; Deng, N. Quantitation of Hydroxyl Radicals from Photolysis of Fe(III)-Citrate Complexes in Aerobic Water. *Environ. Sci. Pollut. Res.* **2006**, *13*, 156−160.

(34) Ye, Y.; Jiang, Z.; Xu, Z.; Zhang, X.; Wang, D.; Lv, L.; Pan, B. Efficient removal of Cr(III)-organic complexes from water using UV/Fe(III) system: Negligible Cr(VI) accumulation and mechanism. *Water Res.* **2017**, *126*, 172−178.

(35) Wan, D.; Zuo, J.; Chen, Y.; Chen, Q.; Zuo, Y. Photodegradation of amitriptyline in Fe(III)-citrate-oxalate binary system: Synergistic effect and mechanism. *Chemosphere* **2018**, *210*, 224−231.

(36) Bianco, A.; Passananti, M.; Brigante, M.; Mailhot, G. Photochemistry of the Cloud Aqueous Phase: A Review. *Molecules* **2020**, *25*, 423.

(37) Wang, Z.; Chen, X.; Ji, H.; Ma, W.; Chen, C.; Zhao, J. Photochemical Cycling of Iron Mediated by Dicarboxylates: Special Effect of Malonate. *Environ. Sci. Technol.* **2010**, *44*, 263−268.

(38) Qiu, J.; Ishizuka, S.; Tonokura, K.; Enami, S. Reactions of Criegee Intermediates with Benzoic Acid at the Gas/Liquid Interface. *J. Phys. Chem. A* **2018**, *122*, 6303−6310.

(39) Santos, P. S. M.; Cardoso, H. B.; Rocha-Santos, T. A. P.; Duarte, A. C. Oxidation of benzoic acid from biomass burning in atmospheric waters. *Environ. Pollut.* **2019**, *244*, 693−704.

(40) Deng, Y.; Zhang, K.; Chen, H.; Wu, T.; Krzyaniak, M.; Wellons, A.; Bolla, D.; Douglas, K.; Zuo, Y. Iron-catalyzed photochemical transformation of benzoic acid in atmospheric liquids: Product identification and reaction mechanisms. *Atmos. Environ.* **2006**, *40*, 3665−3676.

(41) Pang, H.; Zhang, Q.; Wang, H.; Cai, D.; Ma, Y.; Li, L.; Li, K.; Lu, X.; Chen, H.; Yang, X.; Chen, J. Photochemical Aging of Guaiacol by Fe(III)−Oxalate Complexes in Atmospheric Aqueous Phase. *Environ. Sci. Technol.* **2019**, *53*, 127−136.

(42) Dong, P.; Chen, Z.; Qin, X.; Gong, Y. Water Significantly Changes the Ring-Cleavage Process During Aqueous Photooxidation of Toluene. *Environ. Sci. Technol.* **2021**, *55*, 16316−16325.

(43) Andreozzi, R.; Marotta, R. Removal of benzoic acid in aqueous solution by Fe(III) homogeneous photocatalysis. *Water Res.* **2004**, *38*, 1225−1236.

(44) Weller, C.; Horn, S.; Herrmann, H. Photolysis of Fe(III) carboxylato complexes: Fe(II) quantum yields and reaction mechanisms. *J. Photochem. Photobiol., A* **2013**, *268*, 24−36.

(45) Hug, S. J.; Leupin, O. Iron-Catalyzed Oxidation of Arsenic(III) by Oxygen and by Hydrogen Peroxide: pH-Dependent Formation of Oxidants in the Fenton Reaction. *Environ. Sci. Technol.* **2003**, *37*, 2734−2742.

(46) Voelker, B. M.; Sulzberger, B. Effects of Fulvic Acid on Fe(II) Oxidation by Hydrogen Peroxide. *Environ. Sci. Technol.* **1996**, *30*, 1106−1114.
(47) Harris, E.; Sinha, B.; van Pinxteren, D.; Tilgner, A.; Fomba, K. W.; Schneider, J.; Roth, A.; Gnauk, T.; Fahlbusch, B.; Mertes, S.; Lee, T.; Collett, J.; Foley, S.; Borrmann, S.; Hoppe, P.; Herrmann, H. Enhanced Role of Transition Metal Ion Catalysis During In-Cloud Oxidation of SO_2. *Science* **2013**, *340*, 727−730.
(48) Zhang, F.; Yu, X.; Sui, X.; Chen, J.; Zhu, Z.; Yu, X.-Y. Evolution of aqSOA from the Air−Liquid Interfacial Photochemistry of Glyoxal and Hydroxyl Radicals. *Environ. Sci. Technol.* **2019**, *53*, 10236−10245.
(49) Gen, M.; Zhang, R.; Chan, C. K. Nitrite/Nitrous Acid Generation from the Reaction of Nitrate and Fe(II) Promoted by Photolysis of Iron−Organic Complexes. *Environ. Sci. Technol.* **2021**, *55*, 15715−15723.
(50) Weller, C.; Tilgner, A.; Bräuer, P.; Herrmann, H. Modeling the Impact of Iron−Carboxylate Photochemistry on Radical Budget and Carboxylate Degradation in Cloud Droplets and Particles. *Environ. Sci. Technol.* **2014**, *48*, 5652−5659.
(51) Corral Arroyo, P.; David, G.; Alpert, P. A.; Parmentier, E. A.; Ammann, M.; Signorell, R. Amplification of light within aerosol particles accelerates in-particle photochemistry. *Science* **2022**, *376*, 293−296.
(52) Huang, K.-H.; Wei, Z.; Cooks, R. G. Accelerated reactions of amines with carbon dioxide driven by superacid at the microdroplet interface. *Chem. Sci.* **2021**, *12*, 2242−2250.
(53) Yan, X.; Bain, R. M.; Cooks, R. G. Organic Reactions in Microdroplets: Reaction Acceleration Revealed by Mass Spectrometry. *Angew. Chem., Int. Ed.* **2016**, *55*, 12960−12972.

https://doi.org/10.1021/acs.est.2c07897
Environ. Sci. Technol. 2023, 57, 4434−4442